

Article

A Comprehensive Analysis of Early Alzheimer Disease Detection from 3D sMRI Images Using Deep Learning Frameworks

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Abstract: Accurate diagnosis of Alzheimer’s Disease (AD) has largely focused on its later stages, often overlooking the critical need for early detection of Early Mild Cognitive Impairment (EMCI). Early detection is essential for potentially reducing mortality rates; however, distinguishing EMCI from Normal Cognitive (NC) individuals is challenging due to similarities in their brain patterns. To address this, we have developed a subject-level 3D-CNN architecture enhanced by preprocessing techniques to improve classification accuracy between these groups. Our experiments utilized structural Magnetic Resonance Imaging (sMRI) data from the Alzheimer’s Disease Neuroimaging Initiative (ADNI) dataset, specifically the ADNI3 collection. We included 446 subjects from the baseline and year 1 phases, comprising 164 individuals diagnosed with EMCI and 282 individuals with NC. When evaluated using 4-fold stratified cross-validation, our model achieved a validation AUC of 91.5%. On the test set, it attained an accuracy of 81.80% along with a recall of 82.50%, precision of 81.80%, and specificity of 80.50%, effectively distinguishing between the NC and EMCI groups. Additionally, a gradient class activation map was employed to highlight key regions influencing model predictions. In comparative evaluations against pretrained models and existing literature, our approach demonstrated decent performance in early AD detection.

Keywords: early Alzheimer’s disease detection; 3D-CNN; deep learning; sMRI images; explainability



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1. Introduction

The human brain features a highly complex anatomical structure, making it exceedingly difficult to diagnose and analyze various brain disorders such as tumors, Alzheimer’s disease, and headaches and to develop effective treatments for these conditions [1]. Magnetic Resonance Imaging (MRI) has become a crucial tool for distinguishing these disorders from normal brain patterns, and is used extensively for clinical diagnosis [2,3]. Although MRI is widely used for detecting brain abnormalities, the analysis of MRI scans is inherently complex and often requires advanced techniques for accurate quantitative assessment. This process is particularly challenging due to the labor-intensive and error-prone nature of manual segmentation [4,5]. Moreover, manual diagnosis becomes progressively more difficult in the early stages of disease when patterns most closely resemble normal brain activity, underscoring the critical need for dependable and advanced automated detection systems [6,7].

Alzheimer’s Disease (AD) is categorized into several stages: Normal Cognition (NC), Early Mild Cognitive Impairment (EMCI), Late Mild Cognitive Impairment (LMCI), and AD [8]. While much research has focused on MCI and the later stages of AD, there is comparatively less emphasis on the precise classification of early-stage conditions such as EMCI [9]. Improving detection and classification at this early stage is crucial for better prevention and intervention, as it holds the potential to significantly impact disease management and patient outcomes [10,11].

Deep Learning (DL) methods are increasingly recognized for enhancing AD diagnosis due to their capability to uncover intricate patterns across multiple layers of data [7,12].

DL techniques have revolutionized performance in various domains, including object recognition, detection, tracking, and image segmentation. Recently, DL models, particularly Convolutional Neural Networks (CNNs), have demonstrated significant success in medical imaging, particularly for organ segmentation and disease detection [13]. These models are effectively utilized to analyze, classify, and detect cognitive states from MRI and multimodal data [14]. Examples of these networks include popular pretrained CNNs such as ResNet (152 layers), GoogLeNet (22 layers), AlexNet (8 layers), and VGG (16–19 layers).

Recent advancements in medical imaging literature have explored the use of 3D-CNN models for classifying three-dimensional images, capitalizing on their ability to fully leverage contextual information [15] to encode the spatial information [16]. Nonetheless, the application of 3D-CNNs has been limited by computational constraints, leading to the use of simpler architectures and smaller image sizes [17,18]. Additionally, detecting AD with 3D-CNNs typically requires inputting either the entire image or specific regions of interest (ROIs). This approach can necessitate training a large number of parameters on a small dataset, which increases the risk of overfitting [16].

Our study aims to develop an efficient 3D-CNN architecture that can effectively analyze the entire sMRI image as input and classify individuals into two distinct groups: EMCI and NC. By addressing the inherent heterogeneity in brain structures and patterns associated with these conditions, we hope that this approach will significantly improve the differentiation of subtle cognitive changes.

While CNN models often encounter challenges related to “black-box” explainability, recent advancements in “explainable” machine learning (ML) techniques such as Local Interpretable Model-agnostic Explanations (LIME) [19], SHapley Additive exPlanations (SHAP) [20], and Gradient-Weighted Class Activation Map (GRAD-CAM) [11] have made significant progress in addressing these issues. In this study, we employ GRAD-CAM to demonstrate how the model generates heat maps that highlight the locations of discriminative regions associated with CNN predictions. This is achieved by computing CAMs using gradient information.

The major contributions of our paper include:

1. **3D Convolution for Volumetric Data:** The proposed model utilizes 3D convolutional layers, making it well suited for analyzing volumetric data such as 3D MRI, where spatial relationships across depth, height, and width are critical.
2. **Patient-Level Prediction with Limited Data:** We have developed a 3D-CNN architecture for patient-level predictions using each patient’s whole 3D MRI scans. This approach is effective even with limited datasets, which is a common challenge in medical imaging.
3. **MaxPooling and Dense Layer Impact:** The use of 3D MaxPooling (with a pool size of $2 \times 2 \times 2$) and smaller dense layers improves the model’s performance and reduces overfitting, outperforming configurations with larger pooling sizes ($3 \times 3 \times 3$).
4. **Kernel Regularization for Overfitting Control:** L2 kernel regularization on convolutional and dense layers helps to control overfitting in deep learning models for high-dimensional 3D data.
5. **Extensive Data Preprocessing:** We applied advanced image preprocessing steps to the 3D images in order to improve the model’s strength, including image registration and skull stripping, as well as SMOTE.
6. **Comparative Analysis:** A comparative analysis with existing techniques for early AD detection, including pretrained networks, underscores both the strengths and limitations of our approach. This analysis not only compares performance in patient-level prediction but also provides visualization aids to improve interpretability.

In Section 2, we review various studies on AD classification, focusing on different ML and DL techniques, particularly those applied to 3D image classification for early AD detection. Section 3 outlines our data collection and preprocessing methods along with the proposed 3D-CNN model. In Section 4, several pretrained CNN models are introduced. Section 5 presents the performance analysis and results for all models. Section 6 offers a comparative analysis, discussing the strengths and limitations of our model. Finally,

Section 7 provide a summary of our research and discusses potential future directions in this field.

2. Related Works

In this section, we provide an overview of some of the previous works on AD detection using ML models. We have summarized these reviewed works in Table 1.

Table 1. Literature review.

Reference	Dataset	Models	Classes	Performance	Limitations
[21], 2024	16,600 MRI brain image slices (Patients: N/A)	Explainable CNN + MKSCDDL	AD, NC, EMCI, LMCI, MCI	Accuracy: 98.27, Sensitivity: 98.87, Specificity = 96.46, AUC= 98.2	Lacks an explanation of validation processes and use of PET modality. Integration of LIME and CAMs is insufficiently detailed.
[17], 2023	817 MRI images (Patients: 188 AD, 400 MCI, 229 NC)	3D-CNN	AD, MCI, NC	Validation AUC: 88, Test AUC: 72 (whole brain)	Performance metrics were limited to AUC.
[2], 2022	379 MRI images (Patients: 174 AD, 205 NC)	3D-CNN (IGnet)	AD, NC	Accuracy: 67.57, Sensitivity: 61.11 (only MRI images)	Focus on latest stage of disease.
[22], 2021	100 MRI images (Patients: 100, 25/classes)	Fisher score + Multitask feature selection + SVM	AD, LMCI, EMCI, NC	Accuracy: 82, Sensitivity: 80, Specificity: 84 (EMCI vs. NC)	Very small sample size.
[4], 2023	3264 MRI images (Patients: N/A)	CNN	Four types of brain tumor images	Accuracy: 93.3, AUC: 98.43, Recall: 91.19, loss: 0.25	-
[11], 2022	7509 MRI image slices (Patients: 125, 25/classes)	ResNet-18 + DenseNet-121 + Randomized weight	AD, MCI, LMCI, EMCI, NC	Accuracy: 98.86, Sensitivity: 98.89, Precision: 98.94 (5Ways)	Slice-level prediction with limited details for the four test samples.
[23], 2020	2400 fMRI-4D (Patients: 120, 30/classes)	3D-CNN	AD, LMCI, EMCI, NC	Accuracy: 93, Sensitivity: -, Precision: 93.18	-
[24], 2022	360 MRI + PET images (Patients: 151 EMCI, 209 LMCI)	Hybrid 3D-CNN + generative adversarial network (GAN)	EMCI, LMCI	Accuracy: 83.78, Sensitivity: 80.95, Specificity: 87.50, AUC: 88.99	Subjective preprocessing method

5Ways: AD/ MCI/ LMCI/ EMCI/ NC.

Recent advancements in artificial intelligent have significantly improved the detection and confirmation of AD. In this evolving domain, various ML/DL approaches offer distinct advantages and limitations. Adrash et al. [21] developed a medical image classification framework that integrates advanced CNNs, kernel methods, and explanation algorithms to achieve high accuracy and interpretability. Their method employs a custom CNN with a Multi-feature Kernel Supervised within-class-similar Discriminative Dictionary Learning (MKSCDDL) and interpretable models such as LIME and CAMs to categorize images into five classes: AD, NC, MCI, LMCI, and EMCI. This approach ensures both precise and interpretable results. Post-classification, scandent decision trees and transfer learning techniques further refine accuracy and generalizability, achieving a 10-fold cross-validated

accuracy of 90.25% for EMCI classification and an overall accuracy of 98.27% with an AUC of 98.2% for all classes. While effective, this study does not adequately address the integration of multimodal datasets such as Positron emission tomography (PET) images. The study used 12,000 labeled MRI slices for training and a separate set of 4600 slices for testing. While 10-fold cross-validation was applied, the results were not clearly explained. Additionally, the emphasis on slice-level assessments restricts the model's capacity for detailed individual evaluations.

Chakraborty et al. [17] proposed a 3D-CNN model for classifying and extracting features from 3D MRI images. Their study involved classifying 817 participants from the ADNI-1 dataset into three groups: 188 with AD, 400 with MCI, and 229 with NC. The authors explored both whole-brain 3D images and segmented subparts, arguing that 2D models would limit information extraction and hinder the ability to learn more robust features. However, using 3D-CNNs presents challenges such as increased data handling complexity and higher GPU usage during model optimization. Their CNN model achieved a validation AUC of 88% and an average test AUC of 72% across ten repeated runs for classification using the entire brain structure, demonstrating the model's effectiveness in feature extraction without relying on predefined ROIs from existing software. Additionally, the authors compared their 3D-CNN model to pretrained networks such as AlexNet, ResNet50, and VGG16. Despite this, they did not observe significant improvements in testing accuracy with these pretrained models. Their multibranch CNN model trained on subparts of 3D image patches provided an increase in prediction accuracy, yielding a validation AUC of 86% and test AUC of 76%.

Wang et al. [2] utilized a subset of the ADNI-1 dataset comprising 379 participants (AD patients or NC) to develop IGnet, a deep learning approach that integrates 3D imaging and genetic data for automated classification. The IGnet-I model, which relies solely on 3D brain MRI data, achieved an accuracy of 67.57% and an AUC of 78%. However, the model that integrated both imaging and genetic data achieved a higher accuracy of 83.78% and an AUC of 92%, demonstrating the effectiveness of multi-input models in providing more comprehensive information for disease classification. This underscores the advantage of combining imaging and genetic information for AD classification. The authors suggested further investigation into whether 3D ResNet could enhance classification performance even further. The IGnet model employed a 3D-CNN architecture designed for preprocessed 3D brain MRI data. Due to GPU memory limitations and sample size constraints, the network featured five convolutional layers and five pooling layers with filter sizes of 4, 8, 16, 32, and 32, respectively. This configuration effectively balances depth and computational efficiency. As noted by [25,26], 3D-CNNs are highly effective at leveraging spatial contextual information through 3D convolution and pooling for enhanced detection and prediction accuracy. Although this work on 3D networks is insightful, it mainly addresses the late stages of AD.

Sheng et al. [22] analyzed brain imaging and genetic data from 100 individuals (25 from each group) to diagnose various cognitive conditions: (1) NC vs. EMCI, (2) NC vs. LMCI, (3) NC vs. AD, (4) EMCI vs. LMCI, (5) EMCI vs. AD, and (6) LMCI vs. AD. They used a feature selection process to identify five imaging features and five genetic features. Their approach achieved high classification accuracy, reaching 98% for NC vs. AD and 82% for NC vs. EMCI using both genetic and sMRI data as well as sMRI data alone. Classification performance was evaluated using linear SVM after applying Fisher scores and multi-task feature selection to both genetic and imaging data. While the results are promising considering the small sample size, the lack of improvement in accuracy with the addition of genetic data may not be solely due to the model reaching a saturation point with sMRI images or to the genetic data introducing noise. Instead, this issue could stem from the extremely limited dataset of only 25 images per cohort and the way in which the model integrates additional information. Therefore, the authors' explanation for this finding requires further validation, particularly given the potentially critical role of genetic data in the EMCI stage [27].

Odusami et al. [11] included 125 participants, with 25 individuals assigned to each of the AD, MCI, EMCI, LMCI, and NC categories. To simultaneously extract features from MRI images, they utilized two pretrained models: ResNet-18 and DenseNet-121. These features were then concatenated to facilitate AD classification. To address inconsistencies within the fully connected layer, they proposed employing a weight randomization technique such as Kaiming or Xavier to reduce the disparity between the final feature maps generated by the two networks. The weighting mechanism is intended to effectively capture useful features from both models.

The study evaluated the models' performance on a dataset of 7509 MRI images, using 5256 images for training and 2253 for validation. Although they reported a notable accuracy of 93.06% in their multiclass classification utilizing the categories of AD, NC, EMCI, and LMCI, their claim of achieving an accuracy "much higher" than the 93% noted by Parmar et al. [23] needs to be substantiated. The authors identified several potential avenues for future research, including the use of multimodal inputs to better differentiate between the EMCI and LMCI groups, training a baseline model solely on brain images, and leveraging a larger dataset to address potential overfitting issues.

The authors employed GRAD-CAM on sagittal views of MRI images, demonstrating that White Matter Hyperintensities (WMH) are associated with prediction of the EMCI class. Additionally, they identified involvement of the vermis of the cerebellum and the fourth ventricle in predicting AD, while the internal cerebral vein was linked to predictions for NC and MCI, as dilation of the cerebral basal vein correlates with the volume of the white matter hippocampus. In a separate study by Jin et al. [24] which employed GRAD-CAM [28] to visualize MRI images in sagittal, axial, and coronal views, the hippocampus and temporal lobe were found to be associated with the progression of AD. Similarly, Velazquez et al. [29] employed this method on Diffusion Tensor Imaging (DTI) images to generate pixel heatmaps for two groups of EMCI patients, namely, those who progressed to AD and those who did not.

Mahmud et al. [4] proposed a CNN architecture designed for the efficient early detection of brain tumors from MR images, emphasizing the superiority of MRI in tumor detection. Their dataset comprised four types of brain tumor images: meningioma (937 images), no tumor (500 images), pituitary tumor (900 images), and glioma tumor (926 images). The study compared their CNN model against several established architectures, including ResNet-50, VGG16, and Inception V3. For training and evaluation, they used 80% of the data for training and split the remaining 20% equally between validation and testing. In addition, they employed augmentation techniques such as rotation, width and height shifting, and zooming to enhance the dataset and improve model performance.

With a total dataset of 3264 MR images, their CNN model achieved an accuracy of 93.3%, AUC of 98.43%, recall of 91.19%, and loss of 0.25. When compared to other models, their CNN demonstrated superior performance: ResNet-50 achieved accuracy, AUC, and loss values of 81%, 94%, and 0.85, respectively; VGG16 yielded 71.60%, 89.90%, and 1.18; and Inception V3 reached 80%, 89.14%, and 3.67. These results highlight the effectiveness of their proposed model over the alternatives. However, the study was limited in its ability to visualize critical brain tumor areas, largely due to the absence of post hoc explanation tools.

In a manner similar to the aforementioned study [4] but applied to 3D sMRI images, we evaluated our CNN architecture against several established models. Our research specifically targeted the classification of EMCI vs. NC. This classification is crucial for identifying individuals with significant cognitive heterogeneity compared to those with normal cognition. Our CNN model outperformed transfer learning models on this classification task, demonstrating superior results in distinguishing these two very similar brain conditions.

3. Methodology

3.1. Proposed Architecture

In our study, an input image with a size of $128 \times 128 \times 128$ voxels was sent to an initial convolutional layer with 16 filters. A $128 \times 128 \times 128 \times 16$ feature map and a kernel size of $5 \times 5 \times 5$ was used. The convolutional layer's output was then forwarded to a max-pooling layer feature map of $64 \times 64 \times 64 \times 16$, decreasing the size of the spatial data for the subsequent layer by half. This result was then fed to a further convolutional layer with filter values of 16 and a $64 \times 64 \times 64 \times 16$ feature map with a $3 \times 3 \times 3$ kernel size. After that, the output was forwarded to the max-pooling layer feature map of $32 \times 32 \times 32 \times 16$ to cut the amount of spatial data for the next layer in half. Another convolutional layer and another pooling layer came next. The feature map of $32 \times 32 \times 32 \times 32$ in size was made up of 32 filter values and a kernel size of $3 \times 3 \times 3$ in the final convolutional layer, while the final pooling layer had a feature map of $16 \times 16 \times 16 \times 32$. Throughout these steps, the padding was consistently set to 'same', ensuring that the spatial dimensions were preserved as needed during the training process. The newly created 320-dimensional fully connected dense layer received the flattened final output of the previous convolutional layer. The proposed 3D-CNN architecture's configuration is depicted in Figure 1. The model was trained, validated, and tested using 70 epochs, a batch size of 16, and a learning rate of 0.0001. Along with the Adam optimizer, a categorical cross-entropy-based loss function was calculated to find the loss value. To address overfitting, we applied L2 kernel regularization with a strength of 0.01 to each convolutional layer, along with two dropout layers (rates of 0.3 and 0.2) following each dense layer. Additionally, we used L2 kernel regularization with strengths of 0.0001 and 0.001 on the consecutive dense layers. These values were determined to be optimal during fine-tuning, considering the max pooling and overall model structure. This approach helped to stabilize the validation loss and accuracy, contributing to more consistent performance across epochs and folds.

The methodology consisted of several key stages. Initially, we gathered data from an online source (adni.loni.usc.edu) and proceeded with data preprocessing. Using a field strength of 3 Tesla, T1-weighted sagittal acquisition, 3D type, and matrix Z from 176 to 211, we retrieved 446 3D images for the EMCI and NC classes from the ADNI3 dataset. However, later attempts to replicate the same groups with identical criteria revealed discrepancies in subject counts, likely due to variations in the dataset. We then trained our images using various machine learning models, for which the dataset was divided into three groups: 80% for training and validation, and 20% for testing. During the validation stage, we employed a stratified K-fold cross validation system, focusing on two types of brain images associated with the EMCI and NC groups. To evaluate our findings, we utilized multiple metrics, including accuracy, recall, AUC, and loss.

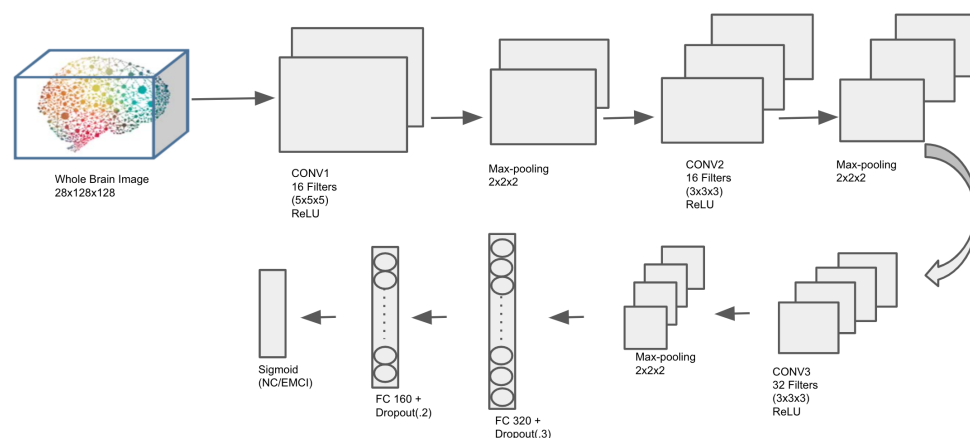


Figure 1. Convolution neural network architecture (3D-CNN) used for the whole-brain structure.

3.2. Environment Setup

We configured our GPU-accelerated TensorFlow environment using Anaconda on an Alienware Aurora R13 computer with an NVIDIA GeForce RTX 3090, which has a compute capability of 8.6. We utilized a Jupyter Notebook for developing our Python code. Additionally, we implemented TensorFlow's mixed-precision policy with compute capability of at least 7.0 to enable mixed-precision training, which optimizes performance and reduces memory usage by using both 16-bit and 32-bit floating-point types.

3.3. Dataset Collection

This study utilized data from the ADNI database (adni.loni.usc.edu), which was launched in 2003 as a public-private partnership led by Principal Investigator Michael W. Weiner, MD. The primary goal of ADNI has been to test whether MRI, PET, other biological markers, and clinical and neuropsychological assessment can be combined to measure the progression of MCI and early Alzheimer's disease.

The ADNI database comprises five phases: ADNI1, ADNI-GO, ADNI2, ADNI3, and ADNI4. ADNI3, initiated in 2016, includes cases of EMCI available for researchers to analyze. We utilized these data with a selection of T1-weighted 3D MRI scans collected during initial and subsequent visit for 446 subjects (282 NC and 164 EMCI), each with corresponding T1-weighted sMRI images in NIfTI format and clinical demographic details in CSV files. Detailed demographic information is provided in Table 2.

Table 2. Subjects' demographic information.

	EMCI	NC
Number	164	282
Gender (F/M)	66/98	142/140
Age-mean [range]	76.20 [62.7–91.1]	78.75 [57.6–95.4]
Weight-mean [range]	78.12 [35.8–112.0]	75.2 [40.4–135.2]

3.4. Dataset Preprocessing

Preprocessing is a vital stage that prepares data for effective training. MR images sourced from patient databases often exhibit low clarity and quality. To enhance our images, we employed image registration and skull stripping techniques. The process utilized the FMRIB Software Library (FSL) along with the MNI152 standard template for registration [30]. This process aligns neuroimaging modalities to a consistent fixed-size template. Skull stripping further improves segmentation outcomes, especially in CNN classification, by removing non-brain structures which introduce noise, such as the skull and scalp [31]. These techniques are commonly employed in neuroimaging research to enhance analytical accuracy and to facilitate comparisons across various studies. We also identified several images from dataset that were completely black and needed to be removed. Additionally, we addressed class imbalance by applying the Synthetic Minority Oversampling Technique (SMOTE), which analyzes the feature space of minority class data points and utilizes their k-nearest neighbors to generate synthetic samples. For visualization of the preprocessing steps applied to each image, starting with registration on the left, followed by skull stripping in the middle, and the finalized image after preprocessing on the right, an example is provided in Figure 2. The images were viewed using FSLeyes, an interactive image viewer for visualizing neuroimaging analysis results, which is part of the FSL suite developed by Jenkinson et al. [32].

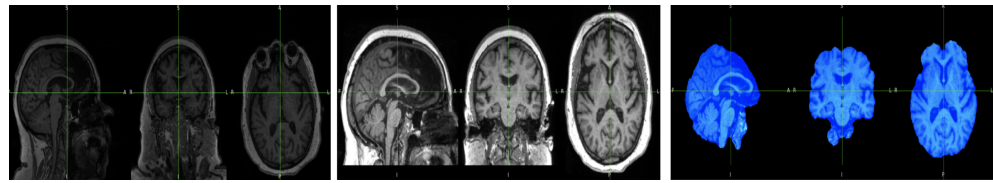


Figure 2. Registration, skull stripping, and final image in FSLeves (left to right).

3.5. Validation Process

Our initial dataset consisted of 446 3D MRI scans, with one image per patient in Nifti format. After removing black images and applying SMOTE for data augmentation, the dataset was expanded to 550 images. To evaluate the model's performance, we utilized a 4-fold stratified cross-validation strategy. In this approach, the dataset was divided into four folds; three were used for training, while one served as the validation set. This process was repeated such that each fold was used as the validation set one time.

To assess the model's ability to generalize, we reserved 20% of the entire image set as an independent testing set that was not used during training or validation. This testing set allowed us to evaluate the model's performance on new unseen data. Moreover, because we did not provide a fixed seed for our results, the test set differed with each kernel restart, consequently remaining entirely distinct from the training and validation sets.

The model was trained on the combined training and validation sets. After training, its performance was assessed on the testing set to provide an estimate of how well it might perform on future data. The benefit of using 80% of the data for training and validation is that it provides the model with a larger dataset to learn from, potentially improving its ability to generalize. However, it is important to note that although the testing set is reserved for performance evaluation, it may not fully represent the entire dataset, meaning that the performance estimates could be subject to bias. Following our attempt to fine-tune the model based on the combined training and validation sets, we also ran separate experiments using a training–testing split to train the model on a larger dataset. In this case, 80% of the data were used only for training, with 20% reserved for testing. The results from this analysis are presented in Section 5.2.

3.6. Performance Metrics

To evaluate the ML models and analyze their performances, we considered the metrics of accuracy, recall, and area under the curve (AUC).

3.6.1. Accuracy

Accuracy measures the number of correct predictions divided by the total number of samples. The accuracy can be calculated by applying Equation (1), where TP stands for True Positive, TN for True Negative, FN for False Negative, and FP for False Positive:

$$Accuracy = ((TN + TP) / (TP + TN + FP + FN)) \times 100\%. \quad (1)$$

3.6.2. Recall

Recall is one of the other most important metrics for evaluating ML models. The recall can be calculated as follows:

$$Recall = TP / (TP + FN). \quad (2)$$

3.6.3. Area Under the Curve

The AUC evaluates how effectively the model distinguishes between positive and negative categories. Higher AUC values indicate better performance on the part of the model.

4. Transfer Learning Models

Transfer learning is an ML approach that draws on knowledge from a related domain to enhance performance in a target domain [33]. This technique is commonly applied to pretrained neural networks such as AlexNet, ResNet-50, and VGG16 [34]. Rather than starting from scratch, an endeavor that can be time-intensive and demands substantial GPU resources and large image datasets, these models are initially trained on source domains (e.g., ImageNet) and subsequently fine-tuned using data from the target domain. However, despite its innovative nature, transfer learning may introduce discrepancies between general image features and the specific features relevant to neurodegenerative diseases [21].

AlexNet is a pioneering CNN architecture developed by Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton which won the ImageNet Large-Scale Visual Recognition Challenge (ILSVRC) in 2012 [35]. The AlexNet architecture features eight layers comprising multiple convolutional and pooling layers, which significantly improves image classification performance and has laid the groundwork for future advancements in DL and computer vision. Figure 3 shows the AlexNet architecture. The Residual Network architecture (ResNet) [36], which won first prize at ILSVRC-2015, is designed around the concept of residual functions. The short form, ResNet-50, features a series of convolutional units that utilize average pooling. Unlike typical neural networks, where the output of each layer is fed directly into the next, ResNet introduces skip connections that allow for more effective training as the number of parameters and layers increases. As illustrated in Figure 4, these residual blocks help to mitigate the vanishing gradient problem, enhancing the network's performance and stability during training.

In this paper, we utilize the three-dimensional versions of these models and present comparisons based on various metrics. Table 3 summarizes the metrics of accuracy, recall, specificity, and precision.

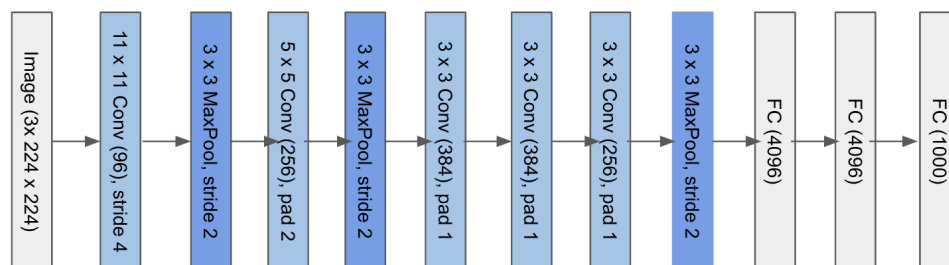


Figure 3. Architecture of 2D AlexNet.

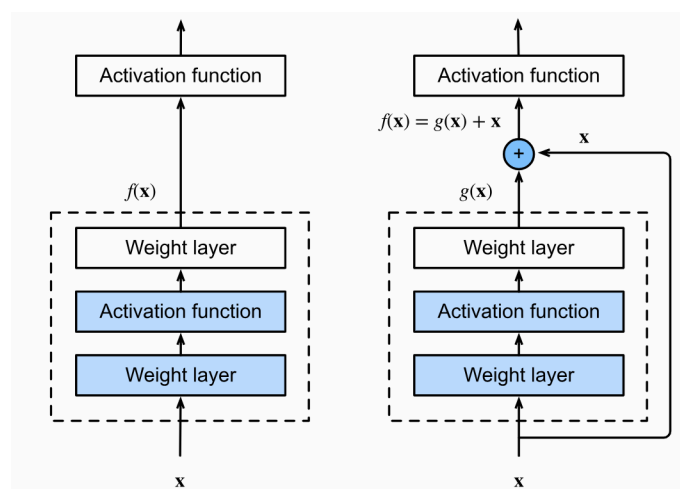


Figure 4. Regular block vs. residual block.

Table 3. Performance comparison of different models under identical conditions.

Models	Optimizer	Epochs	Accuracy (%)	Specificity (%)	Recall (%)	Precision (%)
AlexNet	Adam	70	69.62	77.45	61.82	75.34
ResNet-50	Adam	70	57.46	26.55	88.36	56.00
Developed 3D-CNN	Adam	70	81.80	80.50	82.50	81.80

AlexNet and ResNet-50 in 3D format.

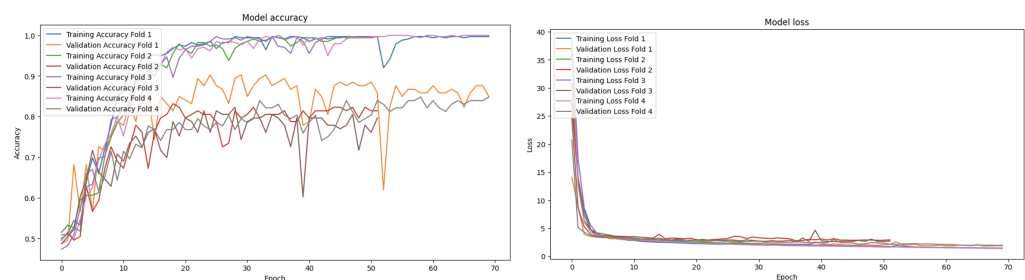
5. Results Analysis

5.1. Experiment 1

In this section, we describe the implementation of transfer learning models such as AlexNet and ResNet-50 in our study. These models have been widely used in similar research for comparisons with CNN architectures, as demonstrated in [4,17]. Using 3D brain sMRI images from the ADNI dataset, we focused on distinguishing between the EMCI and NC groups to enable early disease detection. This requires a finely-tuned algorithm that can identify subtle abnormalities, especially given the limited EMCI dataset. Table 3 summarizes the average test performance of the models over ten runs validated across four folds, with comparative results illustrated in the subsequent figures. All models were trained under identical conditions, including the number of epochs, optimizers, learning rates, and data splitting. The only difference between the models was in their respective CNN architectures.

After the preprocessing steps, we started our analysis with 550 3D images for training, validation, and testing. The models were trained without a fixed seed to ensure that their predictions on the test set would closely align with real-world scenarios, thereby enhancing the generalizability of the resulting models. Figure 5 shows the training and validation accuracy results across four folds, resulting in 330 training, 110 validation, and 110 test images. As illustrated, both the training and validation accuracy increase steadily, while the loss decreases. Despite incorporating regularizers and dropout layers, we faced challenges with overfitting, likely due to the dataset's limited size.

We also performed 9-fold cross-validation while reserving 10% of the dataset for testing, resulting in 440 training, 55 validation, and 55 test images. This significantly improved validation and test accuracy by increasing the training dataset from 330 images in the previous 4-fold setup. However, GPU limitations led to our code halting during the sixth fold. Despite efforts to mitigate this issue, including adjusting the batch size, modifying input sizes, clearing the Keras backend session, and implementing mixed precision training, none of these strategies successfully resolved the interruption.

**Figure 5.** Proposed 3D-CNN model's accuracy and loss.

Given that ResNet-50 showed the lowest performance across all metrics, we opted to include only the results from AlexNet for visual comparison with our model.

To better visualize the training and validation accuracy and loss across four folds, we present the averages in Figures 6 and 7. While both models show normal average accuracy trends, the significant discrepancy between the average training and validation loss in the AlexNet model compared to our model indicates that our model performs substantially better. This difference is less apparent when examining only the average training graphs in Figure 6.

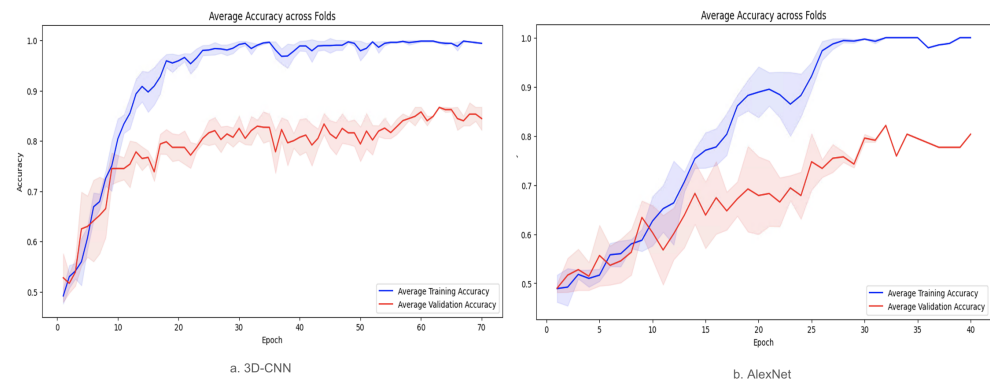


Figure 6. Aggregated accuracy across four folds.

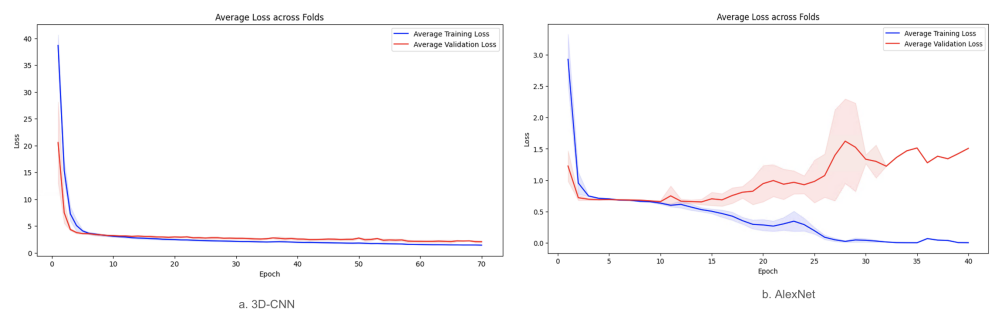


Figure 7. Aggregated loss across four folds.

Figure 8 presents the AUC curves for the 3D-CNN and AlexNet models. The AUC metric illustrates each model's ability to distinguish between classes across different folds. Each curve represents the AUC score plotted against the false positive rate, with higher AUC indicating superior performance. Our 3D-CNN model achieved an impressive average validation AUC of 91.5%, outperforming AlexNet, which had an average validation AUC of 78%. This demonstrates the consistent superiority of 3D-CNN in classification accuracy across various thresholds.

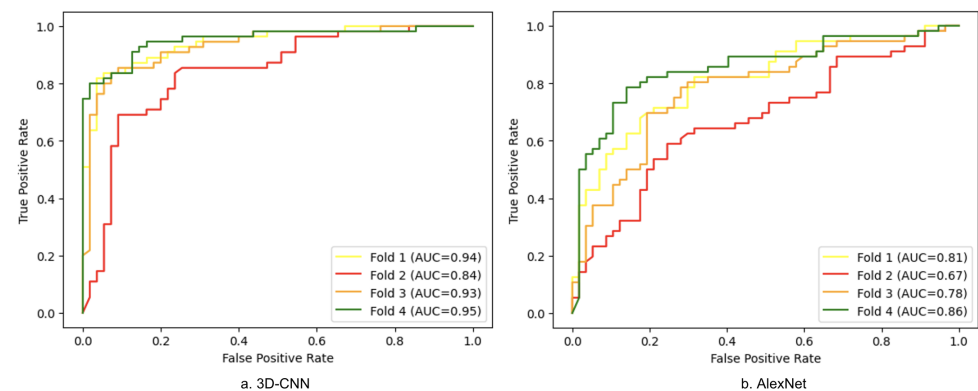


Figure 8. ROC curve results.

Additionally, we visualized the Grad-CAM results for our proposed model on the EMCI and NC classes, as shown in Figure 9. Key regions that are critical for predicting the EMCI class include the hippocampus, temporal lobe, precuneus, and parts of the cingulate gyrus, especially visualized in sagittal views. These areas are associated with Alzheimer's progression, as highlighted in [24]; however, it is important to note that several runs produced inaccurate heatmaps due to key factors such as small group sizes, which limited the model's learning capacity and led to overfitting. Additionally, the model's high nonlinearity due to not using a fixed seed during training and to the inherent randomness

of the optimization algorithm contribute to gradient issues that affect the clarity of the resulting heatmap. Expanding the training dataset could help to address these challenges and improve the model's visualization and generalizability in future work.

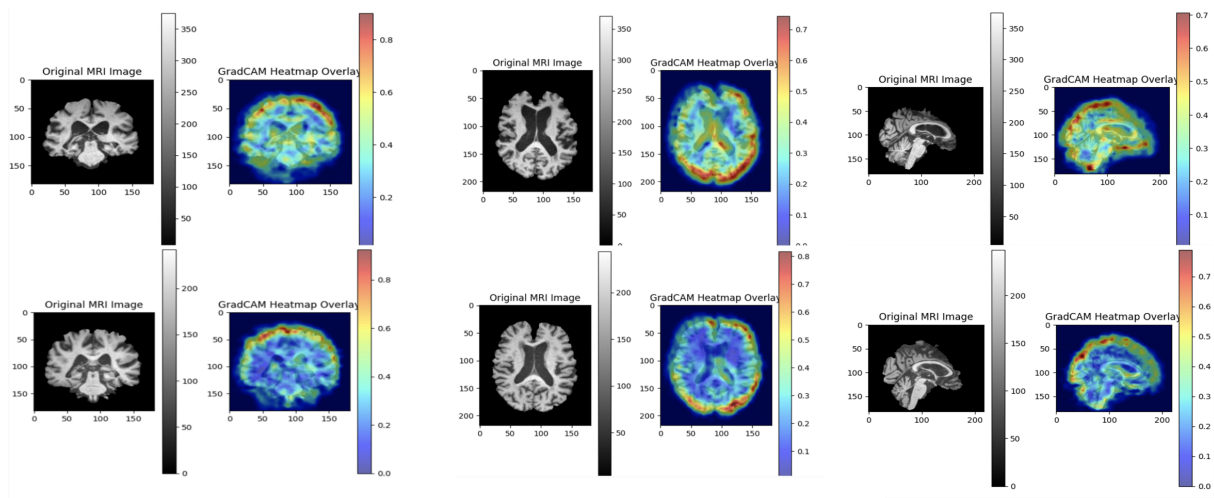


Figure 9. Coronal, axial, and sagittal views of the Grad-CAM heatmaps for EMCI (top) and NC (bottom).

5.2. Experiment 2

To evaluate our model with a larger training set, we conducted fine-tuning based on Experiment 1, then ran a separate test using only the training and test sets (excluding the validation set) under the same conditions. We used the same 3D images but allocated 80% of the data for training and 20% for testing. As in the first experiment, we report the average test performance over ten runs. The model training results for both conditions are presented in Table 4.

Table 4. Performance comparison across the two experimental setups

Models	Optimizer	Epochs	Accuracy (%)	Specificity (%)	Recall (%)	Precision (%)
Developed 3D-CNN (Exp1)	Adam	70	81.80	80.50	82.50	81.80
Developed 3D-CNN (Exp2)	Adam	70	91.24	91.58	90.88	92.26

Exp1: training, validation, and test splits; Exp2: training and test splits.

As shown in Table 4, Experiment 2 outperformed Experiment 1, likely due to the model being trained on a larger training dataset. However, in practical settings results are typically presented and compared using training, validation, and test sets. Therefore, while we report the results from both Experiment 1 and Experiment 2 in Table 4, we only present the figures for Experiment 1 here in order to facilitate a more direct comparison with state-of-the-art models that follow similar training and testing protocols.

6. Discussion

We compared our proposed model with a number of state-of-the-art methods, specifically those focused on EMCI vs. NC classification. Although direct comparison with most studies is not entirely feasible, the results presented in Table 5 demonstrate that our method utilizing 3D whole-brain MRI images performs competitively on the EMCI vs. NC classification task. To ensure a fair comparison, we report only the results from Experiment 1, as most previous studies employed separate training, validation, and testing splits for model training.

Compared to the approaches by Sheng et al. [22] and Abbasian et al. [37], both of which combined multiple input types for the same EMCI vs. NC classification task using different methods, our results are achieved using a single MRI modality. This highlights the usefulness of our approach, although a complete comparison is not possible due to the differences in methodology and modality type.

While our performance does not exceed that of Jiang et al. [8], who utilized a VGG16 network, it is important to note that their analysis lacks results from a separate test set, limiting the generalizability of their findings. Additionally, their approach to subject-level classification differs significantly from ours; they trained their CNN model for support vector machine classification using a slice-level dataset, and organized 120 folders from single subjects containing 32 slices each. This 2020 study claimed to be the first to apply CNNs to brain MRI for EMCI vs. NC classification.

Chakraborty et al. [17] reported results using a 3D-CNN model on whole-brain MRI images, achieving a training AUC of 88% and a testing AUC of 72% for classifying AD, MCI, and NC. In comparison, our approach, which also utilizes a 3D-CNN architecture, achieved an average validation AUC of 91.5% for classifying EMCI and NC. Although our results outperform theirs in terms of AUC, a direct comparison is not entirely appropriate, as their model aimed to classify more advanced stages of AD. However, given that we focused on the earlier stages of AD where cognitive patterns are often very similar to normal cognition, our results are promising.

Moreover, key differences emerge when comparing our work to that of El-Assy et al. [38], who trained their models using at least 1296 2D MRI scans. Notably, their study focused on a five-way multiclass classification approach, while we concentrated on per-patient binary classification with 3D images. Similarly, direct comparison with Ali et al. [39] is complicated by their use of a different dataset (Kaggle) and 2D MRI scans, which limits direct comparisons between the two studies.

Our proposed method offers several advantages over other methods for early AD detection:

1. Our method emphasizes 3D image classification, which has been shown to significantly enhance detection accuracy over traditional 2D approaches, particularly in capturing spatial relationships within brain images.
2. Unlike methods that rely on image count, our approach operates at the patient level, minimizing image overlap across datasets and ensuring more robust evaluation. It is also compatible with hybrid settings, incorporating multiple input analyses for improved accuracy based on more comprehensive subject data.
3. We conducted two experiments, one to validate the model using standard training-validation-testing splits and another to increase the amount of training data by using only the training and testing split after validating the first experiment. This resulted in a more thorough training process with more data.
4. Our approach is one of the first to use a 3D-CNN to generate heatmaps that extract intricate features from MRI brain images at the very early stages of AD. This approach can increase the interpretability of the resulting model.
5. We trained our CNN model on a relatively small dataset, which required extensive fine-tuning to prevent overfitting. The results demonstrate the model's ability to adapt to limited data while maintaining accuracy.
6. Our results are provided without fixing the random seed for different model runs, allowing for increased model variability and generalization while avoiding over-reliance on specific initializations. Additionally, we reserved an untouched randomly-selected test set apart from the k-fold cross-validation to ensure reliable final results.

Our suggested approach does have some limitations. First, our method relies solely on the ADNI dataset, raising concerns about its generalizability to other databases. Different imaging setups in other datasets may require additional preprocessing steps and could affect model performance. Second, while our approach provides heatmaps, there are instances where heatmap generation fails, likely due to overfitting, which could lead to

issues such as vanishing gradients. Further work is needed to improve model explainability and mitigate these issues. Third, there is a lack of studies focusing exclusively on the comparison between the EMCI and NC groups, which makes it challenging to directly compare our results with the existing literature in this specific context.

Table 5. Comparison of the proposed approach with recent research.

Study	Datbase	Method	Subjects	Accuracy	Specificity	Recall	Modality
[22], 2022	ADNI	Multitask feature selection SVM	25EMCI, 25NC	82%	84%	80%	2DMRI+SNP
[40], 2018	ADNI	Ensemble learning	42EMCI, 42NC	81%	78.57%	83.3%	2DMRI
[37], 2023	ADNI	Hybrid-CNN	164EMCI, 282NC	85%	81%	87%	2DMRI+DI
[17], 2023	ADNI	3D-CNN	188AD, 400MCI, 229NC	-	-	-	3DMRI
[8], 2020	ADNI	VGG16 (Finetuning)	70EMCI, 50NC	89.4%	89.3%	92.9%	2DMRI
[38], 2024	ADNI	CNN	171AD, 72LMCI, 233MCI, 240EMCI, 580NC	99.30%	-	99.30%	2DMRI
[39], 2024	Kaggle	CNN	2560Healthy, 2561dementia	98.24%	-	-	2DMRI
Ours (Exp1)	ADNI	3D-CNN	164EMCI, 282NC	81.80%	80.50%	82.50%	3DMRI

DI: demographic information, SNP: single nucleotide polymorphism, SVM: support vector machine.

In conclusion, our proposed strategy not only aids in the early detection of AD at the patient level but also provides valuable insights by visually tracking extracted features from the deep decision layers of the CNN model through GRAD-CAM. This capability enhances the explainability of the model, making our findings more transparent, trustworthy, and applicable in real-world clinical settings.

7. Conclusions and Future Work

Early identification of mild cognitive impairment is crucial for preventing progression to AD later in life. However, detecting the earliest stage is challenging because its brain patterns, cognitive tests, and physiological assessments often closely resemble those of Normal Cognition (NC). Although identifying progressive stages of cognitive decline and associated brain patterns with high accuracy can be easier, there is a risk of patients not receiving appropriate treatment due to the already developed disease condition.

While MR imaging has significantly aided in classifying brain diseases, precise detection of the earliest stages of AD remains challenging. To address this, we have developed a 3D-CNN model that utilizes 3D MRI from EMCI and NC patients. Our model aims to achieve efficiency through metrics such as accuracy, recall, and precision. We also evaluated other pretrained CNN models to compare the outcomes. Despite the limitation of a small EMCI dataset within the ADNI repository and the need for more substantial GPU resources for 3D image analysis, our model demonstrates superior performance compared to the AlexNet and ResNet-50 models, as shown in the results section.

Given the structure of our subject-level model, it has the potential to be evolved into a hybrid model that incorporates multiple modalities, which could allow for integration of complementary information to enhance early AD diagnosis. Future work could further improve the model in this paper by integrating a hybrid CNN architecture that includes additional inputs such as Genome-Wide Association Studies (GWAS) to better identify genetic variants associated with brain patterns. Because the earliest stages of AD are influenced by genetic factors, such approaches could uncover new biomarkers and improve early detection and classification. Additionally, this research could aid the neuroimaging genetics community in exploring the connections between genetics and brain features.

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